Power: grid, solar or both

How the farm should get its electricity, what each option costs, and the steps to put it in, for the owner and the electrician.

Status: advice of 2026-10-28. Loads are estimates [E] until the galley's cooking fuel and the guest-room heaters are chosen; prices are as published on the dates given.

Recommendation, in one line: connect to the grid, add a solar system of about 10 kWp with about 15 kWh of batteries for the things that must never stop, heat water with the sun, and keep a generator only as an emergency backup.

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What the farm uses

Every building, with the energy it uses on a normal day and the most it draws at once. The method: each appliance's power times the hours it runs, then a diversity factor for things that never all run together [E].

Building	Each day (kWh)	Peak (kW)	Note
Studios, two	16	13.8	30 A each at 230 V (D-021); hob, 30 L heater, AC provision
Guest rooms, three	12	9	each with its own shower (D-029): a heater each, or solar hot water
Galley	25	14	electric cooking; about 8 kWh if it cooks on gas
Bar	5	2	fridges, lights, music
Study room	7	2	12 laptops, the screen, fans
Office, public bathroom, guest kitchen	9	5	
Pumps: borehole and booster	5	2	run at midday, when the sun is strongest
Outdoor lights, Wi-Fi, CCTV	5	1	the things that must never stop

Building	Each day (kWh)	Peak (kW)	Note
Farm total	about 84	about 40	with diversity; about 65 kWh a day if the galley cooks on gas

The facts that decide it

Fact	Value	Source
A new connection within 37 m of the line	56,000 RWF, single or three phase, meter included	REG connection policy [V ENG-18]
From 37 m to 800 m	56,000 RWF plus the cost of the line	REG [V ENG-18, ENG-19]
Further, or more power than the line carries	a medium-voltage line and a transformer, paid by the customer, owned by REG	REG [V ENG-19]
Longest allowed wait	5 working days, 2 months with an LV extension, 3 months with an MV extension	RURA [V ENG-22]
Price of grid power, business	376 RWF/kWh above 100 kWh a month, plus 18% VAT: about 444	RURA, from 1 October 2025 [V ENG-20, ENG-21]
Price of grid power, hotels	239 RWF/kWh below 660,000 kWh a year, plus VAT: about 282	RURA [V ENG-20]; ask REG whether the farm qualifies
Outages	about 17 a year, 11 hours in all, nationally	World Bank [V ENG-23]; no Bugesera figure found
The line at Gashora	the Kanazi medium-voltage feeder	REG outage notice [V ENG-29]
Sun at Gashora	4.35 kWh a day from each kWp of panels, about 3.9 in November and December	Global Solar Atlas [V ENG-24]
Tax on solar equipment	no VAT on panels, inverters, deep-cycle batteries, solar water heaters and pumps until 30 June 2028	RRA [V ENG-25]
Licence	none for a system under 50 kW used on site	RURA [V ENG-26]
Selling surplus to the grid	not possible yet: net metering is only being explored	[V ENG-27]

Fact	Value	Source
Diesel	about 2,927 RWF a litre	June 2026 [V ENG-28, secondary]

The three options

	Grid only	Solar only (off-grid)	Grid with solar (hybrid)
What you build	the connection, a site board and cables to each building	panels for the worst month, large batteries, a generator for cloudy spells	the connection, about 10 kWp of panels, about 15 kWh of batteries
Cost to start	lowest if the line is near	highest: panels and batteries for 84 kWh a day	in between
Running cost	every kWh at about 444 RWF	almost none, but batteries wear out	the sun covers the day; the grid covers the rest
When the grid fails	everything stops	nothing changes	the batteries keep Wi-Fi, CCTV, fridges, pumps and some lights going
Fits this farm when	the line is close and money is short	the line is far and the quote is very high	the line is within reach: the likely case

What solar saves: every kWp whose power is used on the farm produces about 1,589 kWh a year, worth about 705,000 RWF at the business price, or about 448,000 at the hotel price [E, from ENG-20, ENG-24]. With no net metering, size the panels to what the farm uses in daylight, not more.

The recommendation

1. **Grid first.** Ask for a three-phase, non-residential connection; the poles already run along the road (photo of 2026-10-28). Ask in writing whether the hotel tariff applies.
2. **Solar for the day.** About 10 kWp of panels on a hybrid inverter set to zero export: they cover the pumps, the hot water, the study room and lunch in the galley.
3. **Batteries for the night essentials.** About 15 kWh of LiFePO4 batteries for Wi-Fi, CCTV, fridges and freezers, the pumps and the path lights, so an outage goes unnoticed.
4. **Sun for hot water.** Solar water heaters on the guest rooms and the public bathroom; the studios keep their 30 L electric boxes (D-024).

5. **Generator as a spare only.** A small diesel set for long outages, wired so it can feed the site board; never the main supply.
6. **Run the big loads at midday.** Pump, heat water and wash in daylight, when the panels make free power.

Steps

1. Take ID, the land title and the construction permit to the EUCL Bugesera branch and ask for a written connection quote: three phase, about 40 kW, the distance from the nearest line to the gate.
2. Decide the galley's cooking fuel (gas or electric): it moves the daily total by about 17 kWh.
3. Ask three RURA-licensed installers (class D covers solar) to quote a 10 kWp hybrid system with 15 kWh of LiFePO4 batteries, zero export, under 50 kW, VAT-exempt.
4. Ask the tax authority (RRA) in writing to confirm that the LiFePO4 batteries count as deep-cycle, so they stay VAT-free.
5. The electrician designs the site board: one 32 A feeder for each studio, feeders for every other building, the inverter, the generator change-over.
6. Choose with the quotes in hand, using the triggers below, and record the decision in the decision log.

What would change the advice

- The connection quote is far higher than the cost of the extra panels and batteries for going off-grid: choose solar only.
- Net metering comes into force: add more panels and sell the surplus.
- The galley cooks on gas: the hybrid system can shrink, to about 8 kWp.
- Outages at the farm turn out longer than the national figure: add battery capacity.